Internal Practical : Date : 22/10/26

FCFS

```c
#include <stdio.h>

int main()
{
    int n, i, j;
    int total_wt = 0, total_tat = 0;
    float avg_wt, avg_tat;

    printf("24CS099 - Hetvi Taank\n");

    printf("Enter number of processes: ");
    scanf("%d", &n);

    int at[n], bt[n], ct[n], wt[n], tat[n];

    for(i = 0; i < n; i++)
    {
        printf("\nP%d Arrival Time: ", i + 1);
        scanf("%d", &at[i]);

        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);
    }

    for(i = 0; i < n - 1; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(at[i] > at[j])
            {
                int temp;

                temp = at[i];
                at[i] = at[j];
                at[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
```

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    }

    ct[0] = at[0] + bt[0];

    for(i = 1; i < n; i++)
    {
        if(at[i] > ct[i - 1])
            ct[i] = at[i] + bt[i];
        else
            ct[i] = ct[i - 1] + bt[i];
    }

    for(i = 0; i < n; i++)
    {
        tat[i] = ct[i] - at[i];
        wt[i] = tat[i] - bt[i];

        total_wt += wt[i];
        total_tat += tat[i];
    }

    avg_wt = (float)total_wt / n;
    avg_tat = (float)total_tat / n;

    printf("\nProcess\tAT\tBT\tCT\tWT\tTAT\n");

    for(i = 0; i < n; i++)
    {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
               i + 1, at[i], bt[i], ct[i], wt[i], tat[i]);
    }

    printf("\nAverage Waiting Time = %.2f", avg_wt);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat);

    return 0;
}
```

Output screenshot

